Android Dedicated (COSU) / Kiosk Device Enrollment & Management in Intune

Implementation Solution Guide

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Executive Summary — Android Dedicated (COSU) / Kiosk Device Enrollment & Management (Microsoft Intune)

Frontline, warehouse, retail, logistics, and clinical devices exist to do **one job** — scan, sign, display, capture — and nothing else. Left as ordinary tablets they wander into personal use, get set up inconsistently, and are hard to lock down and scale. The fix is Android Enterprise's **dedicated (COSU)** model in **Microsoft Intune**: company-owned, user-less devices that **boot straight into a kiosk** and stay locked to their allowed app(s).

This solution stands that up end to end: the **dedicated enrollment profile**, enrollment by **token / QR / NFC / zero-touch**, the **kiosk** (single-app or multi-app), **device restrictions**, **managed Google Play** app delivery to devices that have no signed-in user, and **remote management** — all validated on a pilot before you scale. It is a complete, **deploy-it-today** package written for an administrator with no prior dedicated-device experience.

What you get

• A plain-language **Solution Overview** of the dedicated/COSU model, the four enrollment methods (token/QR/NFC/zero-touch), single- vs multi-app kiosk, and how apps reach a user-less device.

• A **Requirements Guide** with verify / implement-if-missing for every prerequisite (the category Android requirements — Intune tenant, managed Google Play binding, roles, network, pilot — are referenced, not repeated).

• An **Implementation Guide** of precise, ordered **Intune admin center + Microsoft Graph** steps: enrollment profile, dynamic device group, app assignment, the kiosk profile, enrollment, and validation — pilot-first, each artifact named, with pass/fail gates and rollback.

• Five **PowerShell scripts** for readiness discovery, safe configuration (`-WhatIf`), the only-allowed-apps kiosk validation, fleet health/inventory reporting, and scope targeting.

• **Testing** and **Troubleshooting** guides for a confident rollout.

What this delivers

Purpose-built Android devices **enrolled at scale** by the method that fits the fleet, **locked to their app(s)** by single- or multi-app kiosk, **fed managed Google Play apps** to the device with no user, and **remotely managed** — consistent and tamper-resistant. OEM/rugged devices (Zebra, Samsung) are supported at the pattern level via OEMConfig.

The deployment model

Assess → **Design** → **Configure** → **Pilot** → **Validate** → **Expand** — build the profile and kiosk, prove them on a handful of devices, then roll out in waves.

An honest note

Dedicated is for **corporate-owned, single-purpose** devices only; reversing the enrollment model requires a factory reset. The **AMAPI transition and whether a change forces re-enrollment are actively evolving** — where behavior is version-dependent, this package flags **"verify against current Microsoft guidance / your tenant"** rather than asserting it.

Outcome

A fleet of Android dedicated (COSU) devices that boot straight to the job, enrolled consistently, locked to their apps, and remotely managed — validated before you scale.

1. Solution Overview — Android Dedicated (COSU) / Kiosk Device Enrollment & Management in Intune

What this solution is

This solution stands up **Android Enterprise dedicated devices** — Google's **COSU** (Corporate-Owned, Single-Use) model — in **Microsoft Intune**. Dedicated devices are company-owned, have **no user affinity**, and boot straight into a **kiosk**: either **one app** (single-app kiosk) or a **curated set of apps** on a locked launcher (multi-app kiosk). This is the model for frontline, warehouse, retail, logistics, and clinical devices that exist to do one job and nothing else.

It covers the whole path: choosing and building the **dedicated enrollment profile**, enrolling devices by **token / QR code / NFC / zero-touch**, applying the **kiosk profile and device restrictions**, delivering **managed Google Play** apps to devices that have no user signed in, and **remotely managing** the fleet — validated on a pilot before you scale.

It is **vendor-neutral**. Samsung Knox and Zebra (and other rugged/kiosk OEMs) are covered at the **pattern level** — where an OEM extension or OEMConfig app changes a step, this solution names the action and defers the OEM console/agent specifics to the vendor's documentation.

The dedicated (COSU) model in one paragraph

A dedicated device is enrolled as an Android Enterprise **device-owner** with **no user account**. Because there is no user, everything is targeted at the **device**, not a person: enrollment is driven by an **enrollment profile / token**, and policy and apps are assigned to a **device group** (typically a **dynamic device group keyed on the enrollment profile name**). The device runs in **kiosk mode** — Android's lock-task mode — so the user cannot leave the allowed app(s), reach settings, or use the device personally.

The four enrollment methods (pick per device fleet)

The dedicated enrollment profile produces a **token** (a string) plus a **QR code** that carry the enrollment. From a factory-reset device you choose how that token is delivered:

- **QR code** — on the setup "Welcome" screen, tap the screen several times to launch the QR reader,

then scan the profile's QR code. The most common method for dedicated devices; no Google account or typing needed.

- **Token / afw#setup (manual)** — enter `afw#setup` where the setup asks to sign in, then supply the

enrollment token. Useful where a camera/QR is impractical.

- **NFC** — bump a factory-reset device against an NFC "programmer" device running an NFC-provisioning

app that carries the token. A bulk method for identical hardware; the NFC bump payload is deferred to the OEM/tooling.

- **Zero-touch enrollment** — devices bought through a **zero-touch reseller** are pre-assigned an

enrollment configuration, so they enroll automatically on first boot with no scanning. Requires the **Android zero-touch portal** (reseller-provisioned) linked to your EMM; the portal specifics are deferred to Google/reseller documentation.

All four land the device on the **same** dedicated enrollment profile and therefore the same kiosk and restriction policy — the method only changes how the token reaches the device.

Single-app vs multi-app kiosk

- **Single-app kiosk** — the device locks to **one** app (e.g. a scanning app, a survey app, a

digital-signage player). The app launches on boot and the user cannot leave it. Configured on the Android Enterprise **device restrictions** profile by setting the kiosk to that one app.

- **Multi-app kiosk** — the device shows a **locked launcher** with a **curated set** of apps and

nothing else. Intune delivers this using Microsoft's **Managed Home Screen** app (`com.microsoft.launcher.enterprise`) as the launcher, with the allowed apps listed in the kiosk configuration. You control the grid, whether the user can change Wi-Fi/Bluetooth/volume, a session PIN, screen-saver, and more.

Both are built on the `androidDeviceOwnerGeneralDeviceConfiguration` device-restrictions profile; multi-app additionally requires the Managed Home Screen app deployed to the device group.

Managed Google Play app delivery to devices with no user

Because a dedicated device has **no signed-in user**, apps are delivered to the **device**: managed Google Play apps (`androidManagedStoreApp`) are assigned to the **dedicated device group** as **Required**, so they install at/after enrollment without any user action. The kiosk allow-list must reference apps that are actually **assigned and installed** — an app named in the kiosk but not assigned to the group will not appear. Keeping the **app assignments and the kiosk allow-list in sync** is a first-class validation in this solution.

Remote management

Dedicated devices are managed remotely through Intune device actions — **restart, wipe, retire, sync, remote-lock**, and (where supported) **remote assistance** — plus policy/app updates that flow to the device group. The Reporting script surfaces fleet health (compliance, last check-in, OS version, OEM model) so you can spot stale or drifted devices.

What is authored exactly vs deferred

- **Exact navigation is given** for the **Intune admin center** (`intune.microsoft.com`) — the

dedicated enrollment profile, the kiosk/device-restrictions profile, managed Google Play app assignment, enrollment restrictions, and device actions — and for **Microsoft Graph** automation.

• **The managed Google Play binding** (account enterprise vs Google Workspace) is a **category prerequisite** (global requirement G2) — confirm it is in place before enrolling.

• **OEM specifics** (Samsung Knox, Zebra) and the **zero-touch reseller portal** are integrated at the **pattern level**; their consoles/agents are deferred to the vendor's documentation.

• **The AMAPI transition and whether a policy change forces re-enrollment are actively evolving** —

where behavior is version-dependent this solution flags **"verify against current Microsoft guidance / your tenant"** rather than asserting it.

Benefits

• **Boots to the job** — devices power on straight into the allowed app(s); no personal use, no wandering.

• **Consistent at scale** — one enrollment profile + one kiosk policy applied to every device, the same way.

• **User-less by design** — no account to manage; policy and apps target the device group.

• **Choice of enrollment** — token, QR, NFC, or zero-touch, all landing on the same profile.

• **Managed + reversible** — remotely manageable, piloted first, with scripts to validate and report.

Limitations (be honest with buyers)

• **Dedicated = corporate-owned, no user.** It is not for BYOD or personal-plus-work phones (those are other enrollment models). Choose dedicated only for purpose-built devices.

• **Re-enrollment sensitivity.** Changing the **enrollment model** of a device generally requires a **factory reset and re-enrollment**; some kiosk/restriction changes apply live while others may need a reboot. Whether a specific change forces re-enrollment is **version-dependent — verify against current Microsoft guidance / your tenant**, and treat enrollment-model changes as high-impact.

• **OEM/rugged features vary.** Barcode-scanner, OEMConfig, and firmware-update behavior differ by vendor (Zebra, Samsung) and are deferred to the OEM.

• **No managed Google Play binding = no dedicated devices.** The category binding (G2) is a hard prerequisite.

The standard deployment model

> ***Assess the fleet → design the profile + kiosk → configure the enrollment profile, kiosk, and app assignments → pilot-enroll a few devices → validate the locked-down experience → expand in waves.*** *Pattern:* ***Assess → Design → Configure → Pilot → Validate → Expand****, with explicit pass/fail gates and a documented rollback for production changes.*

2. Requirements Guide — Android Dedicated (COSU) / Kiosk Device Enrollment & Management

Verify **every** requirement before configuring. Each entry: **Description · Why required · How to verify · How to implement if missing.** Category-wide Android prerequisites (Intune tenant + Android Enterprise, the **managed Google Play binding**, least-privilege roles, enrollment-method decision + Graph toolchain, network reachability, pilot + change control) are in `global-requirements-android.md` — confirm those **first**; this guide adds the dedicated/COSU specifics and does not duplicate the six.

> ***Object names used throughout (use verbatim):*** *dedicated enrollment profile* `Android Dedicated - COSU - Pilot` *(and* `Android Dedicated - COSU - Prod`*); kiosk / device-restrictions profile* `Android Dedicated - Kiosk - Pilot`*; dedicated device group* `Android Dedicated - Devices - Pilot` *(dynamic, keyed on the enrollment profile name) and* `Android Dedicated - Devices - Prod`*; enrollment-restriction* `Android Dedicated - Enrollment Restriction - Pilot`*; Managed Home Screen app assignment (multi-app only) target = the dedicated device group.*

2.1 Managed Google Play connection in place (category G2)

Description: Intune connected to **managed Google Play** — either the **managed Google Play account enterprise** (tenants without Google Workspace) or the **Google Workspace / Cloud Identity ↔ Intune third-party EMM** binding. One and only one model per tenant. **Why required:** Dedicated enrollment and all app delivery ride on the managed Google Play connection; without it you cannot create Android Enterprise enrollment profiles or assign apps. **Why verify:** Intune → **Devices** → **Android** → **Android enterprise** (or **Enrollment**) shows the connection **bound/active**; run `COSU-Discovery.ps1` (reports the account-enterprise binding). **How to implement if missing:** Establish the binding first (this is a separate category solution), then return here.

2.2 Least-privilege administrative role

Description: Intune Administrator for enrollment-profile, kiosk-policy, and app work (**Global Administrator** only for the initial managed Google Play binding, which is a prerequisite, not part of this solution). **Why required:** Creating enrollment profiles, device-restriction/kiosk profiles, and app assignments requires the corresponding Intune role. **How to verify:** M365 admin center / Entra **Roles** shows Intune Administrator on the operator. **How to implement if missing:** Have a Global Admin assign the least-privilege role under change control.

2.3 Corporate-owned, user-less devices (dedicated is the right model)

Description: The target devices are **company-owned**, purpose-built, and intended to run with **no user affinity** (not personal / not BYOD). **Why required:** Dedicated (COSU) enrolls as an Android Enterprise **device owner with no user**; choosing it for personal devices is the wrong model and cannot be undone without a factory reset. **How to verify:** Confirm ownership and single-purpose use with the device owners; the devices are supplied factory-reset (or will be reset before enrollment). **How to implement if missing:** If devices need a user or personal use, choose a different enrollment model (work profile / fully managed) — not this solution.

2.4 Devices support Android Enterprise device-owner enrollment

Description: The device hardware/OS supports Android Enterprise **device-owner** mode and the chosen enrollment method (QR/NFC readers as needed; camera for QR; NFC for NFC bump). **Why required:** Kiosk/lock-task mode and dedicated enrollment require device-owner mode on a supported Android version; very old or non-GMS devices may not qualify. **How to verify:** Confirm the device model on Android Enterprise's supported/recommended list and the minimum Android version your kiosk features require; a factory-reset test device provisions to device owner. **How to implement if missing:** Use supported hardware; update Android to the required minimum before enrolling.

2.5 The dedicated enrollment profile + enrollment method decision

Description: A decided delivery method — **QR / token (afw#setup) / NFC / zero-touch** — and the Intune **dedicated enrollment profile** `Android Dedicated - COSU - Pilot` that produces the token/QR. **Why required:** Every method lands the device on the same profile; the profile's **token has an expiry** and the QR encodes it. **How to verify:** Intune → **Devices** → **Android** → **Enrollment** → **Corporate-owned dedicated devices** shows the profile with a **non-expired token**; run `COSU-Discovery.ps1` (lists profiles + token expiry + enrolled count). **How to implement if missing:** Create it (Implementation Step 2) or with `COSU-Set-Configuration.ps1`; for **zero-touch** additionally link the reseller-provisioned **zero-touch portal** (deferred to Google/reseller docs).

2.6 Kiosk decision: single-app or multi-app (and the app list)

Description: A decision on **single-app** (one locked app) vs **multi-app** (curated set), and the **exact package name(s)** the kiosk will allow. **Why required:** The kiosk profile is built differently for each, and multi-app additionally needs the **Managed Home Screen** launcher; the allow-list must reference **real, assigned** apps. **How to verify:** You have the app package name(s) (from managed Google Play) and, for multi-app, have identified the **Managed Home Screen** app (`com.microsoft.launcher.enterprise`). **How to implement if missing:** Approve/select the apps in managed Google Play (2.7) and record their package names before building the kiosk.

2.7 Managed Google Play apps approved and available

Description: The kiosk app(s) are **approved in managed Google Play** and available in Intune as `androidManagedStoreApp` entries, ready to assign to the device group. **Why required:** Dedicated devices have no user, so apps must be assigned to the **device** as **Required**; a kiosk app that is not assigned/installed will not appear on the device. **How to verify:** Intune → **Apps** → **Android** lists the app(s); run `COSU-Discovery.ps1` (reports managed store apps). **How to implement if missing:** Approve the app(s) in the managed Google Play surface inside Intune, then assign them in Implementation Step 4.

2.8 A dedicated device group (dynamic, keyed on the enrollment profile)

Description: An Entra **device group** that captures dedicated devices automatically — `Android Dedicated - Devices - Pilot`, typically a **dynamic device group** with a membership rule on the **enrollment profile name**. **Why required:** Dedicated devices have no user, so kiosk policy and apps are assigned to a **device group**; a dynamic rule on the profile name means every device enrolled with the profile is captured with no manual step. **How to verify:** Entra → **Groups** shows the group; run `COSU-Report.ps1` (membership count). **How to implement if missing:** Create it with `COSU-Set-ScopeTargeting.ps1` (default `-WhatIf`; add `-Apply`). Example dynamic rule:

`(device.enrollmentProfileName -eq "Android Dedicated - COSU - Pilot")`.

2.9 Enrollment restriction allowing corporate-owned dedicated Android

Description: Device platform **enrollment restrictions** that **allow** Android Enterprise corporate-owned (dedicated) enrollment for the operators/scope doing the pilot. **Why required:** If a restriction blocks corporate-owned Android or the platform, enrollment fails even with a valid token. **How to verify:** Intune → **Devices** → **Enrollment** → **Enrollment restrictions** → **Device platform restrictions**; confirm Android (corporate-owned) is allowed; `COSU-Discovery.ps1` flags a blocking default. **How to implement if missing:** Adjust/scoped-create the restriction (Implementation Step 1).

2.10 Network reachability for enrollment + managed Google Play (category G5)

Description: The device network can reach the **Google (managed Google Play, Android provisioning)** and **Microsoft (Intune, Entra)** endpoints; QR/token/NFC/zero-touch provisioning all download the DPC and register the device. **Why required:** Blocked endpoints are the most common cause of "provisioning failed / can't complete setup" on dedicated devices. **How to verify:** Confirm the documented Google/Microsoft endpoints are allowed on the device VLAN/SSID. **How to implement if missing:** Allow the endpoints before enrolling (see category G5).

2.11 Pilot scope + change control (category G6)

Description: A small **pilot set** of representative dedicated devices and a **change-control** window; enrollment-model and connector changes are high-impact. **Why required:** The pilot proves the profile + kiosk before fleet rollout and protects production devices; some changes may force **re-enrollment** (verify against current guidance). **How to verify:** A named pilot device set and an approval window exist. **How to implement if missing:** Define the pilot and agree change control before production changes.

Checklist

- Managed Google Play connection bound/active (2.1 / category G2)
- Intune Administrator role on the operator (2.2)
- Devices are corporate-owned, user-less, purpose-built (2.3)
- Hardware/OS supports Android Enterprise device-owner + chosen method (2.4)
- Dedicated enrollment profile `Android Dedicated - COSU - Pilot` with a non-expired token (2.5)
- Single-app vs multi-app decided; kiosk app package name(s) recorded (2.6)
- Kiosk app(s) approved in managed Google Play and present in Intune (2.7)
- Dynamic device group `Android Dedicated - Devices - Pilot` keyed on the profile name (2.8)
- Enrollment restriction allows corporate-owned dedicated Android (2.9)
- Device network reaches Google/Microsoft endpoints (2.10 / category G5)

• Pilot device set + change-control window agreed (2.11 / category G6)

Global Requirements — Microsoft Intune Android (Android Enterprise) Solutions

These requirements apply to **every** solution in the **Android** category (Android Enterprise connection & managed Google Play, Google Workspace ↔ Intune EMM integration, the Android Management API (AMAPI) migration, dedicated/COSU & work-profile enrollment, app management, and enrollment troubleshooting). Verify each before starting a solution's implementation steps.

> ***Vendor boundary (applies to every Android solution).*** *The **Microsoft Intune admin center** (intune.microsoft.com), **Microsoft Graph**, the **managed Google Play** enrollment surface, and the **Google Admin console** (for Google Workspace / Cloud Identity integration) are authored at exact, current click-by-click / cmdlet detail from Microsoft's and Google's public documentation. The **Android Management API (AMAPI)** transition (Intune's move off the legacy Play EMM API) and **web-based / account-driven enrollment** are **actively evolving** — steps, defaults, and whether a change triggers **re-enrollment** are **cited or flagged "verify against current Microsoft guidance / your tenant"**, never asserted from memory or invented. **OEM specifics** (Samsung Knox, Zebra, and other rugged/kiosk vendors) and **third-party apps** are integrated at the pattern level; their consoles are deferred to the vendor's docs. Automation is **PowerShell** (Microsoft Graph) — never with hardcoded credentials or secrets.*

1. Microsoft 365 tenant with Intune and Android Enterprise

• **Description:** A Microsoft 365 tenant with **Microsoft Intune** and Android Enterprise available, and the licensing the capability requires.

• **Why required:** All Android policy, enrollment, and app configuration is performed in Intune.

• **How to verify:** Sign in to the Intune admin center; confirm Android platform availability.

• **How to implement if missing:** Provision/assign Intune licensing before starting.

2. The managed Google Play binding (account enterprise OR Google Workspace)

• **Description:** Intune connected to **managed Google Play** — either the **managed Google Play account enterprise **(for tenants WITHOUT Google Workspace) or the** Google Workspace / Cloud Identity ↔ Intune third-party EMM** integration (for tenants that use Google Workspace). One and only one model applies per tenant.

• **Why required:** Android Enterprise enrollment and app delivery depend on the managed Google Play connection; the two binding models are mutually exclusive and must be chosen deliberately.

• **How to verify:** Intune admin center → **Devices > Android > Android enterprise (or Enrollment)** — confirm the connection status and which model is in place.

• **How to implement if missing:** Establish the correct binding (this is often the first solution in the

line) before enrolling devices.

3. Administrative access (least-privilege Intune roles)

• **Description:** Access with the least-privilege role the solution names — typically **Intune

Administrator**, with** Global Administrator** only where genuinely required (e.g. the initial managed Google Play / Google Workspace binding).

• **Why required:** Enrollment, connector, and app changes require the corresponding role.

• **How to verify:** Confirm the role in the M365 admin center / Entra roles.

• **How to implement if missing:** Have a Global Admin assign the least-privilege role under change control.

4. Enrollment method decision and Graph/PowerShell toolchain

• **Description:** A decided **enrollment method** — personally-owned **work profile** (BYOD), **fully

managed**,** corporate-owned work profile**, or** dedicated / COSU (kiosk) **— plus a workstation with** Microsoft Graph PowerShell** for automation.

• **Why required:** Profiles, restrictions, and app assignment differ by enrollment method; the provided

scripts run through Graph.

• **How to verify:** Confirm the target enrollment method(s) with the device owners; `Connect-MgGraph`

succeeds.

• **How to implement if missing:** Choose the enrollment method and install Graph PowerShell before starting.

5. Network reachability to Google & Microsoft endpoints

• **Description:** Devices and the enrollment path can reach the required **Google (managed Google Play,

Android) **and** Microsoft (Intune, Entra)** service endpoints (no proxy/filter blocks the enrollment or account sign-in).

• **Why required:** Enrollment, managed Google Play account provisioning, and adding a Google account all

require reachability; blocked endpoints are a common cause of "can't add a Google account" failures.

• **How to verify:** Confirm the required endpoints are allowed on the device network.

• **How to implement if missing:** Allow the documented Google/Microsoft endpoints before enrolling.

6. A pilot scope and change control

• **Description:** A **pilot group / small device set** to validate on before fleet-wide rollout, and a

change-control process for production changes (enrollment-model, connector, and AMAPI-migration changes are high-impact and may affect existing devices).

- **Why required:** Every solution follows an **Assess** → **Design** → **Configure** → **Pilot** → **Validate** → **Expand**

model; the pilot proves the design and protects the production fleet.

- **How to verify:** Confirm a named pilot scope and an approval/change window are available.
- **How to implement if missing:** Create a pilot scope and agree change control before production changes.

*The deployment model across every Android solution is the same: **Assess** → **Design** → **Configure** → **Pilot** → **Validate** → **Expand**, with explicit pass/fail gates and a documented rollback for production changes. Automation is delivered as **PowerShell** (Microsoft Graph) — never with hardcoded credentials or secrets. Where the AMAPI transition or enrollment behavior is version-dependent, verify against current Microsoft guidance rather than assuming.*

3. Implementation Guide — Android Dedicated (COSU) / Kiosk Device Enrollment & Management

Follow these steps **in order**. The **Intune admin center** and **Microsoft Graph** steps are given click-by-click / cmdlet-exact. **OEM** specifics (Samsung Knox, Zebra) and the **zero-touch reseller portal** are named and deferred to the vendor's documentation. The **managed Google Play binding** (category G2) is assumed already in place (Requirement 2.1).

Consoles used:

- **Microsoft Intune admin center** — `https://intune.microsoft.com`
- **Managed Google Play** (approve apps) — reached from inside Intune's Android app surface
- **Android zero-touch portal / OEM consoles** — deferred to Google/reseller/OEM docs

> ***Model:*** *Configure the enrollment profile, kiosk, restrictions, and app assignments;* ***pilot-enroll a few devices****; validate the locked-down experience; then expand in waves.* ***Assess*** *→* ***Design*** *→* ***Configure*** *→* ***Pilot*** *→* ***Validate*** *→* ***Expand.***
>
> ***Artifact names (use verbatim):*** *enrollment profile* `Android Dedicated - COSU - Pilot`*; kiosk / device-restrictions profile* `Android Dedicated - Kiosk - Pilot`*; dedicated device group* `Android Dedicated - Devices - Pilot` *(dynamic, keyed on the enrollment profile name); enrollment restriction* `Android Dedicated - Enrollment Restriction - Pilot`*.*
>
> ***Risk + rollback (read first):*** *Dedicated enrollment makes a device a* ***device owner with no user*** *— reversing the* ***enrollment model*** *requires a* ***factory reset and re-enrollment****. Kiosk and restriction changes flow to enrolled devices;* ***whether a specific change forces re-enrollment or a reboot is version-dependent — verify against current Microsoft guidance / your tenant. Rollback:*** *to stop new enrollments,* ***delete/expire the enrollment profile token****; to relax the lockdown on already-enrolled devices,* ***unassign*** `Android Dedicated - Kiosk - Pilot` *from the device group (policy reverts on next check-in); a device already provisioned as dedicated returns to normal only by* ***wipe/retire + factory reset****. Always pilot on a* ***few*** *devices before the fleet.*

Phase 1 — Prepare scope and enrollment restrictions

Step 1 — Confirm/scope the enrollment restriction (allow corporate-owned dedicated Android)

1. Intune → **Devices** → **Enrollment** → **Enrollment restrictions** → **Device platform restrictions**.

2. Confirm **Android Enterprise (corporate-owned)** enrollment is **Allowed** for the operators/scope

doing the pilot. If a default blocks it, **Create restriction** named `Android Dedicated - Enrollment Restriction - Pilot` allowing corporate-owned Android, and assign it to a scope for the pilot.

1. Run `COSU-Discovery.ps1` — it flags a blocking default before you waste a device.

Step 2 — Create the dedicated enrollment profile

1. Intune → **Devices** → **Android** → **Enrollment** → **Corporate-owned dedicated devices** → **Create profile**.

2. **Name** `Android Dedicated - COSU - Pilot`; add a description. Choose **token type** =

Corporate-owned dedicated device (the default dedicated token; if your tenant offers **dedicated with Microsoft Entra shared mode**, that is a distinct option — use plain dedicated for this pilot unless you specifically need shared-mode sign-in).

1. Set a sensible **token expiration** (e.g. 90 days) and **Create**.

2. Open the profile → **Token** to view the **enrollment token string**, the **QR code**, and the

expiry. This one profile feeds **all** enrollment methods (QR / afw#setup / NFC / zero-touch).

> *Prefer automation?* `COSU-Set-Configuration.ps1` *creates the enrollment profile (and the kiosk profile below) via Graph — default* `-WhatIf`*; add* `-Apply`*.*

Step 3 — Create the dedicated device group (dynamic, keyed on the profile)

1. Entra → **Groups** → **New group** → **Security**, **Membership type = Dynamic Device**; name it

`Android Dedicated - Devices - Pilot`.

1. **Dynamic rule:** `(device.enrollmentProfileName -eq "Android Dedicated - COSU - Pilot")`. **Create.**

2. Every device enrolled with the profile is captured automatically — this is how user-less devices are

targeted. (Automate with `COSU-Set-ScopeTargeting.ps1`, default `-WhatIf`.)

Phase 2 — Apps and the kiosk

Step 4 — Approve and assign the kiosk app(s) to the device group (Required)

1. Intune → **Apps** → **Android** → **Add** → **Managed Google Play app**; open the managed Google Play surface,

approve each app the kiosk will run, then **Sync**.

1. For each app → **Assignments** → **Required** → **Add group** = `Android Dedicated - Devices - Pilot`.

Because the device has no user, **Required-to-device** is what installs the app.

1. **Multi-app kiosk only:** also add and assign Microsoft's **Managed Home Screen** app (`com.microsoft.launcher.enterprise`) as **Required** to the same group — it is the locked launcher.

1. Run `COSU-Discovery.ps1` to confirm the apps are present, and note the exact **package names** (you will list them in the kiosk).

Step 5 — Build the kiosk / device-restrictions profile

1. Intune → **Devices** → **Android** → **Configuration** → **Create** → **New policy** → Platform **Android Enterprise**, Profile type **Device restrictions** (under **Fully Managed, Dedicated, and Corporate-Owned Work Profile**). **Name** `Android Dedicated - Kiosk - Pilot`.

1. Expand **Device experience** (or **Kiosk**, depending on the current portal) and set **Enrollment profile type = Dedicated device**, then choose the kiosk mode:

• **Single-app kiosk: Kiosk mode = Single app**, and select the **one** app (by package name) the device locks to. It launches on boot; the user cannot leave it.

• **Multi-app kiosk: Kiosk mode = Multi-app**, **add** each allowed app (package names from Step 4) to the app list, and set the launcher options — grid size, whether to **lock the home button**, allow **Wi-Fi / Bluetooth / volume / flashlight** toggles, a **session PIN / exit**, and **screen saver**. Multi-app uses the **Managed Home Screen** launcher deployed in Step 4.

1. Set the **device restrictions** that harden the kiosk (as appropriate): block adding accounts, block factory reset by user, block safe boot, block USB file transfer, control system update behavior, set the allowed **input methods**, etc. Keep it minimal for the pilot and tighten later.

1. **Assignments** → **Add group** = `Android Dedicated - Devices - Pilot`. **Create.**

> *The kiosk **allow-list must reference apps that are assigned to the device group** (Step 4). An app named here but not assigned will **not appear**.* `COSU-Validation.ps1` *cross-checks this.*

Step 6 — (Optional) OEM hardening at the pattern level

1. For **Zebra / Samsung** rugged devices, deliver OEM settings via an **OEMConfig** app (approved in managed Google Play) assigned to the device group — barcode-scanner profiles, key remaps, firmware update control, etc.

1. Add the OEMConfig app as **Required** to `Android Dedicated - Devices - Pilot` and configure its **App configuration policy**. The exact schema is the **OEM's** — deferred to the vendor's docs.

Phase 3 — Enroll the pilot devices

Step 7 — Enroll by your chosen method (factory-reset device required)

Pick the method that fits the fleet; all land on `Android Dedicated - COSU - Pilot`.

1. **QR code:** on the factory-reset **Welcome** screen, **tap the screen ~6 times** to start the QR

reader (connect Wi-Fi if prompted), then **scan** the profile's QR code. Provisioning downloads the DPC and enrolls the device as dedicated.

1. **Token (afw#setup):** proceed to the Google sign-in prompt and enter `afw#setup`, then supply the **enrollment token string** when asked.
1. **NFC:** run an NFC-provisioning app on a "programmer" device carrying the token and **bump** the factory-reset device (the NFC payload/tooling is deferred to the OEM).
1. **Zero-touch:** a device purchased through a **zero-touch reseller** and assigned your configuration in the **zero-touch portal** enrolls automatically on first boot — no scanning. (Portal/reseller steps deferred to Google/reseller docs.)
1. Let the device finish provisioning; it should land on the **kiosk** (single app, or the Managed Home Screen with the allowed apps).

Step 8 — Confirm the device landed in scope

1. Intune → **Devices** → **Android** → **Corporate-owned dedicated devices** (or **All devices**) shows the new device; it should be **no user / device owner**.
1. Confirm the device joined `Android Dedicated - Devices - Pilot` (dynamic membership can take a few minutes) so kiosk + apps deploy. Run `COSU-Report.ps1` to see it in inventory.

Phase 4 — Validate and expand

Step 9 — Validate the locked-down experience (pilot gate)

1. On the device, confirm: it **boots to the kiosk**; **only the allowed app(s)** are reachable; settings/other apps are blocked; and it is **remotely manageable** (try **Sync** and a benign device action from Intune).
1. Run `COSU-Discovery.ps1` (readiness), `COSU-Validation.ps1` (kiosk allow-list vs assigned apps; only-allowed-apps report), and `COSU-Report.ps1` (fleet health). Complete the **Testing Guide** pilot gate before expanding.

Step 10 — Expand to production in waves

1. Create the production twins: enrollment profile `Android Dedicated - COSU - Prod`, kiosk `Android Dedicated - Kiosk - Prod`, and device group `Android Dedicated - Devices - Prod` (dynamic rule on the **prod** profile name) — or widen the pilot artifacts, per your change process.
1. Enroll production devices in **waves** by the chosen method; assign the same apps + kiosk to the prod device group.
1. Re-run `COSU-Validation.ps1` and `COSU-Report.ps1` after each wave; validate before widening.
2. **Zero-touch at scale:** have the reseller assign the prod configuration to purchased devices so new

stock enrolls on first boot.

Order of operations

`Enrollment restriction → dedicated enrollment profile → dynamic device group → approve + assign apps (Required-to-device) → kiosk / device-restrictions profile (single or multi-app) → (OEM config) → enroll pilot (QR/token/NFC/zero-touch) → confirm in scope → validate locked-down → expand in waves`

5. Testing Guide — Android Dedicated (COSU) / Kiosk Device Enrollment & Management

Run after the kiosk + apps are assigned and at least one device is enrolled (Implementation Steps 7–9), and again after each expansion wave. Back each test with `COSU-Discovery.ps1` / `COSU-Validation.ps1` / `COSU-Report.ps1`.

Test 1 — Enrollment profile + token are valid

Do: Run `COSU-Discovery.ps1`; check Intune → **Devices** → **Android** → **Corporate-owned dedicated devices**. **Expected:** `Android Dedicated - COSU - Pilot` exists with a **non-expired token**; QR code available.

Test 2 — Device enrolls as dedicated (no user)

Do: Enroll a factory-reset pilot device by the chosen method (QR / afw#setup / NFC / zero-touch); check Intune → **Devices**. **Expected:** The device appears as **corporate-owned dedicated**, **no primary user / device owner**, enrolled with the profile.

Test 3 — Device lands in the dynamic device group

Do: Run `COSU-Report.ps1`; check Entra → **Groups** → `Android Dedicated - Devices - Pilot`. **Expected:** The enrolled device is a member (dynamic rule on the enrollment profile name captured it).

Test 4 — Kiosk apps installed (Required-to-device)

Do: Run `COSU-Validation.ps1`; on the device, look for the app(s). **Expected:** Every managed Google Play app assigned **Required** to the group is **installed**; for multi-app, the **Managed Home Screen** launcher is present.

Test 5 — Device boots to the kiosk (single-app)

Do: Reboot a single-app device. **Expected:** It launches **straight into the one allowed app**; the user cannot exit to a launcher, settings, or any other app.

Test 6 — Multi-app kiosk shows only the allowed set

Do: On a multi-app device, review the Managed Home Screen. **Expected: Only** the curated apps appear; no other apps, no full settings; toggles limited to what the kiosk policy allows (Wi-Fi/Bluetooth/volume as configured); session PIN/exit behaves as set.

Test 7 — Only-allowed-apps report is clean

Do: Run `COSU-Validation.ps1`. **Expected:** The kiosk **allow-list matches the apps assigned** to the device group — no app named in the kiosk is missing an assignment, and no unexpected app is exposed. Any mismatch is listed to fix.

Test 8 — Device restrictions hold

Do: On the device, attempt blocked actions (add a Google account, factory reset from settings, USB file transfer, safe boot) per your policy. **Expected:** Each blocked action is **prevented**; allowed actions still work.

Test 9 — Device is remotely manageable

Do: From Intune, run **Sync** and a benign device action (e.g. **Restart**) on the pilot device. **Expected:** The action reaches the device; `COSU-Report.ps1` shows a recent **last check-in** and the device **compliant** (if a compliance policy is assigned).

Test 10 — Fleet health + OEM/inventory breakdown

Do: Run `COSU-Report.ps1`. **Expected:** A clear count of dedicated devices with compliance state, last check-in, OS version, and manufacturer/model (OEM breakdown, e.g. Zebra/Samsung); no unexpected stale or non-dedicated devices in the group.

Pass/fail gate

Gate	Requirement
Enrollment pass	Tests 1–3 pass (profile/token valid; device enrolls dedicated; joins the group)
Kiosk pass	Tests 4–7 pass (apps installed; boots to kiosk; only allowed apps; report clean)
Lockdown + management pass	Tests 8–10 pass (restrictions hold; remotely manageable; fleet healthy)

Do not expand beyond the pilot until all three gates pass. Re-run Tests 4–10 after each wave.

6. Troubleshooting Guide — Android Dedicated (COSU) / Kiosk Device Enrollment & Management

Symptom → likely root cause → fix. Use `COSU-Discovery.ps1` (readiness), `COSU-Validation.ps1` (only-allowed-apps), and `COSU-Report.ps1` (fleet health) to confirm before and after each fix. Where enrollment/kiosk behavior is **version-dependent, verify against current Microsoft guidance / your tenant** rather than assuming.

A. Enrollment

"Can't set up device / provisioning failed" when scanning the QR (or afw#setup)

- **Managed Google Play not bound.** Dedicated enrollment requires the managed Google Play connection

(category G2). **Fix:** run `COSU-Discovery.ps1`; if the binding is not active, establish it first.

- **Device not factory-reset / already has an account.** Device-owner enrollment needs a **clean**

device with no Google account added. **Fix:** factory-reset and enroll **before** any account is added.

- **Network blocks provisioning.** The setup path downloads the DPC and registers the device. **Fix:**

allow the documented Google/Microsoft endpoints on the device VLAN/SSID (category G5).

- **Token expired.** The enrollment profile's token has a lifetime. **Fix:** `COSU-Discovery.ps1` reports

the expiry; extend/rotate the token in the profile and regenerate the QR.

QR reader won't start on the Welcome screen

- **Wrong gesture / device variant.** The QR reader is launched by **tapping the same spot ~6 times** on

the first setup screen; some OEMs vary. **Fix:** follow the OEM's device-owner QR steps; on older devices use `afw#setup` + token instead.

Enrollment is blocked even with a valid token

- **Enrollment restriction blocks corporate-owned Android. Fix:** Intune → **Devices → Enrollment →**

Enrollment restrictions → Device platform restrictions**; allow Android Enterprise (corporate-owned) for the scope. `COSU-Discovery.ps1` flags a blocking default.

NFC or zero-touch device doesn't auto-enroll

- **NFC payload/tooling issue** — the NFC bump payload is OEM/tooling-specific. **Fix:** defer to the OEM

NFC-provisioning steps and confirm the token in the payload is current.

- **Zero-touch not assigned** — the reseller must assign your **zero-touch configuration** to the device

serials in the **zero-touch portal**. **Fix:** confirm with the reseller (portal deferred to Google/reseller docs); the device must be purchased through a zero-touch reseller.

B. Device group / targeting

Device enrolled but got no kiosk or apps

- **Not in the device group yet.** Kiosk + apps are assigned to `Android Dedicated - Devices - Pilot`;

dynamic membership can take a few minutes. **Fix:** wait, then `COSU-Report.ps1` / check Entra group membership.

- **Dynamic rule doesn't match the profile.** The rule must be `(device.enrollmentProfileName -eq "Android Dedicated - COSU - Pilot")` — exact profile name. **Fix:** run `COSU-Set-ScopeTargeting.ps1`; it flags a rule that doesn't reference the profile.
- **Assigned to a user instead of the device group.** Dedicated devices have **no user**; user-targeted assignments never land. **Fix:** re-target apps/kiosk to the **device** group.

C. Kiosk / apps

A kiosk app is missing on the device

- **App named in the kiosk but not assigned.** The allow-list must reference apps that are **Required to the device group**. Fix:** run `COSU-Validation.ps1` — it lists "in kiosk but not assigned"; assign each as Required, then re-check.
- **App still installing.** Managed Google Play installs can lag enrollment. **Fix:** give it time / force a **Sync**; confirm in `COSU-Report.ps1`.

Multi-app kiosk shows a blank/black home screen

- **Managed Home Screen not deployed.** Multi-app uses `com.microsoft.launcher.enterprise` as the launcher. **Fix:** assign **Managed Home Screen** Required to the device group; `COSU-Validation.ps1` FAILs when it's missing on a multi-app kiosk.

Extra apps are visible / device isn't as locked as expected

- **Assigned-but-not-in-kiosk apps.** An app Required to the group but absent from the allow-list can be reachable. **Fix:** `COSU-Validation.ps1` lists these as WARN — add them to the kiosk or remove the assignment.
- **Restrictions too loose. Fix:** tighten the device-restrictions profile (block add-accounts, factory reset by user, USB transfer, safe boot) and re-test (Testing Test 8).

Single-app kiosk doesn't launch the app on boot

- **Wrong package or app not installed. Fix:** confirm the exact **packageId** in the kiosk matches an installed app (`COSU-Validation.ps1`); correct the package and ensure it's assigned Required.

D. Remote management / health

Device won't take remote actions / looks offline

- **Stale check-in.** The device is likely powered off or off-network. **Fix:** `COSU-Report.ps1` flags stale devices (no check-in within `-StaleDays`); investigate connectivity/power before assuming failure.
- **Retired/wiped.** A device removed from management won't respond. **Fix:** confirm state in Intune;

re-enroll (factory reset) if needed.

A policy/kiosk change didn't apply

• **Waiting on check-in, reboot, or re-enrollment.** Some changes apply on next check-in, some need a reboot, and **enrollment-model changes require a factory reset + re-enrollment**. **Fix:** force **Sync**; reboot; and **verify against current Microsoft guidance / your tenant** whether the specific change requires re-enrollment before rolling it fleet-wide.

E. OEM / rugged (pattern level)

• **Scanner / OEM setting not taking effect.** These are delivered by the **OEM's OEMConfig app + app configuration**, not native kiosk settings.** Fix:** confirm the OEMConfig app is assigned Required to the device group and its configuration is correct; the schema is the OEM's — defer to Zebra/Samsung docs.

Escalation checklist

1. `COSU-Discovery.ps1` — binding active, profile + token valid, apps present, no blocking restriction?
2. `COSU-Set-ScopeTargeting.ps1` — device group exists, rule matches the profile, device is a member?
3. `COSU-Validation.ps1` — kiosk allow-list matches assigned apps (only allowed apps), MHS present for multi-app?
4. `COSU-Report.ps1` — device compliant, checked in recently, expected OEM/OS?
5. Still stuck → capture the enrollment/kiosk symptom and **verify the current Microsoft guidance** for dedicated (COSU) / AMAPI behavior in your tenant before a fleet-wide change.

7. Best Practices — Android Dedicated (COSU) / Kiosk Device Enrollment & Management

Field-proven recommendations for a scalable, tamper-resistant dedicated-device fleet.

Design

• **Pick dedicated for the right reason.** Dedicated = corporate-owned, **no user**, single-purpose. If a device needs a user or personal use, it is a different enrollment model — don't force it.

• **One profile per purpose, not per device.** A single `Android Dedicated - COSU - *` profile plus one kiosk policy should serve a whole class of identical devices. Split profiles only when the kiosk app set or hardware genuinely differs.

• **Target the device, never the (absent) user.** Assign everything — apps and kiosk — to a **dynamic

device group keyed on the enrollment profile name**. This is the whole trick to managing user-less devices at scale and needs no manual grouping.

- **Prefer multi-app only when you need it.** Single-app kiosk is simpler and harder to escape; use multi-app (Managed Home Screen) when the job genuinely needs a curated set.

Enrollment method

- **QR is the default; zero-touch is the endgame.** For existing stock, QR is fastest to pilot. For ongoing purchasing at scale, get the **reseller to assign your zero-touch configuration** so new devices enroll on first boot with no touch.
- **Watch the token expiry.** The enrollment profile token expires; `COSU-Discovery.ps1` reports it. Rotate/extend before it lapses or new enrollments fail.
- **Standardize the reset step.** Enrollment requires a **factory-reset** device; bake "reset → enroll → verify kiosk" into your provisioning runbook so staging is consistent.

Apps and kiosk

- **Keep the kiosk allow-list and app assignments in lockstep.** Every app named in the kiosk must be **Required-to-device** on the group; every Required app that should be user-visible should be in the kiosk. Run `COSU-Validation.ps1` on every change — a drifted allow-list is the #1 "app is missing" cause.
- **Assign apps to the group before (or with) the kiosk.** If the kiosk deploys before the app installs, the tile is empty until the app catches up.
- **Harden with intent, not reflexively.** Block what defeats the purpose (add-accounts, factory reset by user, USB transfer, safe boot). Leave what the job needs (a specific input method, a scanner toggle). Over-blocking creates support tickets.

OEM / rugged

- **Deliver OEM settings via OEMConfig, at the pattern level.** Zebra/Samsung scanner profiles, key remaps, and firmware control belong in the **OEM's OEMConfig app + app configuration**, assigned to the device group — not hand-set on devices. The schema is the OEM's; defer to their docs.
- **Control firmware/OS updates deliberately.** Frontline fleets hate surprise updates mid-shift; set the system-update policy to your maintenance window where the OEM/Android supports it.

Operations

- **Pilot on a few, then wave.** Prove the profile + kiosk on a handful, pass all three testing gates, then expand in waves — validating each wave. Never flip the whole fleet at once.
- **Treat enrollment-model changes as high-impact.** Reversing dedicated needs a **wipe + factory reset**;

some kiosk/restriction changes may need a reboot or (rarely) re-enrollment — **verify against current Microsoft guidance / your tenant** before a fleet-wide change.

• **Monitor for stale devices.** Run `COSU-Report.ps1` on a schedule; a device with an old last-check-in is likely powered off, off-network, or drifted — investigate before it fails silently.

• **Document rollback.** For any production kiosk change, know the revert: unassign the kiosk to relax, delete the token to stop new enrollments, wipe/retire to fully reset a device.

Automation & security

• **Run the scripts read-only first.** `COSU-Discovery` / `COSU-Validation` / `COSU-Report` are read-only; the config helpers default to `-WhatIf` — preview before `-Apply`.

• **Least privilege, no secrets.** Connect with the **least Graph scopes** each task needs; never place credentials, tokens, or the enrollment token string in scripts or source control.

• **Keep the enrollment token secret-ish.** The QR/token enrolls any factory-reset device onto your tenant — treat it as sensitive; rotate it if it leaks.

8. Deliverables Summary — Android Dedicated (COSU) / Kiosk Device Enrollment & Management

Documentation

Document	Purpose
Executive Summary	Business case and what's in the box
1. Solution Overview	The dedicated/COSU model, enrollment methods (QR/token/NFC/zero-touch), single- vs multi-app kiosk, app delivery to user-less devices, OEM at pattern level
2. Requirements Guide	Every dedicated/COSU prerequisite with Verify / Implement-if-missing (category Android GRs referenced, not duplicated)
Global Requirements (Android)	Category-wide Android prerequisites, published once
3. Implementation Guide	Exact, ordered Intune + Graph steps: enrollment profile, dynamic device group, app assignment, kiosk (single/multi), enroll, validate; pilot-first with pass/fail + rollback
5. Testing Guide	Validation procedures with pass/fail gates
6. Troubleshooting Guide	Enrollment, kiosk, app-delivery, and remote-management issues by symptom → root cause

Document	Purpose
7. Best Practices	Field-proven recommendations for a scalable dedicated fleet

PowerShell scripts (Microsoft Graph)

Script	What it does	Safety
`COSU-Discovery.ps1`	Dedicated-device readiness: managed Google Play binding, enrollment profile(s) + token expiry + enrolled count, managed store apps, blocking enrollment restrictions	Read-only
`COSU-Set-Configuration.ps1`	Creates the dedicated enrollment profile and the kiosk / device-restrictions profile (single- or multi-app)	Default `-WhatIf`; `-Apply` to write
`COSU-Validation.ps1`	Kiosk / restriction validation: the kiosk **allow-list vs the apps assigned** to the device group — the "only allowed apps" report, mismatches listed	Read-only
`COSU-Report.ps1`	Enrolled-fleet health/inventory: dedicated device count, compliance, last check-in, OS version, OEM (manufacturer/model) breakdown	Read-only
`COSU-Set-ScopeTargeting.ps1`	Scope/config helper: builds the dynamic device group keyed on the enrollment profile name (and reports kiosk/app assignment targeting)	Default `-WhatIf`; `-Apply` to write

What the buyer receives

- This implementation guide as a branded **Word** document and **PDF**.
- All five **PowerShell scripts** in a single ZIP.
- A repeatable process: **Assess** → **Design** → **Configure** → **Pilot** → **Validate** → **Expand.**

Outcome

A fleet of purpose-built Android **dedicated (COSU)** devices that boot straight to the job — enrolled at scale by token / QR / NFC / zero-touch, locked to their app(s) via single- or multi-app kiosk, fed managed Google Play apps to the device, and remotely managed — consistent, tamper-resistant, and proven on a pilot before you scale.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

COSU-Discovery.ps1

Purpose: Assesses whether the tenant is ready to enroll Android Enterprise DEDICATED (COSU) devices: the managed Google Play account-enterprise binding, the corporate-owned dedicated enrollment profile(s) with token expiry and enrolled-device count, managed Google Play apps available to assign, and any device-platform enrollment restriction that would block corporate-owned Android. Read-only against Microsoft Graph — makes no changes. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\COSU-Discovery.ps1 Requires the Microsoft Graph PowerShell SDK and delegated scopes: DeviceManagementServiceConfig.Read.All, DeviceManagementConfiguration.Read.All, DeviceManagementApps.Read.All. EXPECTED RESULTS The managed Google Play binding state, each dedicated enrollment profile with token expiry + enrolled count, a count of managed store apps, and a flag if a default enrollment restriction blocks corporate-owned Android. OEM console + zero-touch reseller portal are deferred to vendor docs. .NOTES Android category — Android dedicated (COSU)/kiosk. Read-only. AMAPI/re-enrollment behavior: verify against current Microsoft guidance / your tenant. > [CmdletBinding()] param([string]$ProfileNameFilter = 'Android Dedicated - COSU')

How to run: powershell -ExecutionPolicy Bypass -File .\COSU-Discovery.ps1 Requires the Microsoft Graph PowerShell SDK and delegated scopes: DeviceManagementServiceConfig.Read.All, DeviceManagementConfiguration.Read.All, DeviceManagementApps.Read.All. EXPECTED RESULTS The managed Google Play binding state, each dedicated enrollment profile with token expiry + enrolled count, a count of managed store apps, and a flag if a default enrollment restriction blocks corporate-owned Android. OEM console + zero-touch reseller portal are deferred to vendor docs. .NOTES Android category — Android dedicated (COSU)/kiosk. Read-only. AMAPI/re-enrollment behavior: verify against current Microsoft guidance / your tenant. > [CmdletBinding()] param([string]$ProfileNameFilter = 'Android Dedicated - COSU')

COSU-Report.ps1

Purpose: Reports the enrolled Android DEDICATED (COSU) fleet: device count, compliance state, last check-in freshness (flags stale devices), OS version spread, and an OEM breakdown by manufacturer/model (e.g. Zebra, Samsung) — so you can audit fleet health and spot drifted or offline devices. Read-only against Microsoft Graph. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\COSU-Report.ps1 ... -DeviceGroup 'Android Dedicated - Devices - Pilot' -StaleDays 7 Requires the Microsoft Graph PowerShell SDK and delegated scopes: DeviceManagementManagedDevices.Read.All, Group.Read.All. EXPECTED RESULTS A dedicated-device count with compliance/last-check-in/OS/OEM breakdowns and a per-device list; stale devices (no check-in within -StaleDays) are flagged. Read-only. .NOTES Android category — Android dedicated (COSU)/kiosk. Read-only. Enrollment-type enum values may vary by version — verify against your tenant. > [CmdletBinding()] param([string]$DeviceGroup = 'Android Dedicated - Devices - Pilot', [int]$StaleDays = 7)

How to run: powershell -ExecutionPolicy Bypass -File .\COSU-Report.ps1 ... -DeviceGroup 'Android Dedicated - Devices - Pilot' -StaleDays 7 Requires the Microsoft Graph PowerShell SDK and delegated scopes: DeviceManagementManagedDevices.Read.All, Group.Read.All. EXPECTED RESULTS A dedicated-device count with compliance/last-check-in/OS/OEM breakdowns and a per-device list; stale devices (no check-in within -StaleDays) are flagged. Read-only. .NOTES Android category — Android

dedicated (COSU)/kiosk. Read-only. Enrollment-type enum values may vary by version — verify against your tenant. > [CmdletBinding()] param([string]$DeviceGroup = 'Android Dedicated - Devices - Pilot', [int]$StaleDays = 7)

COSU-Set-Configuration.ps1

Purpose: Creates the two Intune objects this solution relies on: (1) the corporate-owned DEDICATED (COSU) Android enrollment profile that produces the enrollment token/QR, and (2) the Android Enterprise device-restrictions profile that configures the KIOSK — single-app (lock to one package) or multi-app (Managed Home Screen with a curated app list). Idempotent: skips objects that already exist by display name. Default -WhatIf; pass -Apply to make changes. HOW TO RUN Preview (default): powershell -ExecutionPolicy Bypass -File .\COSU-Set-Configuration.ps1 Single-app kiosk: ... -Apply -KioskMode Single -KioskApps 'com.vendor.scanapp' Multi-app kiosk: ... -Apply -KioskMode Multi -KioskApps 'com.vendor.a','com.vendor.b' Requires the Microsoft Graph PowerShell SDK and delegated scopes: DeviceManagementServiceConfig.ReadWrite.All, DeviceManagementConfiguration.ReadWrite.All. EXPECTED RESULTS The dedicated enrollment profile and the kiosk device-restrictions profile exist (created if missing). Assign apps (Required-to-device) and the kiosk profile to the device group per the Implementation Guide. Multi-app also requires the Managed Home Screen app assigned to the group. .NOTES Android category — Android dedicated (COSU)/kiosk. Default -WhatIf. Verify the current schema/portal against Microsoft guidance before -Apply. > [CmdletBinding(SupportsShouldProcess=$true)] param([string]$EnrollmentProfileName = 'Android Dedicated - COSU - Pilot', [int]$TokenValidityDays = 90, [string]$KioskProfileName = 'Android Dedicated - Kiosk - Pilot', [ValidateSet('Single','Multi')] [string]$KioskMode = 'Single', [string[]]$KioskApps = @(), [switch]$Apply)

How to run: Preview (default): powershell -ExecutionPolicy Bypass -File .\COSU-Set-Configuration.ps1 Single-app kiosk: ... -Apply -KioskMode Single -KioskApps 'com.vendor.scanapp' Multi-app kiosk: ... -Apply -KioskMode Multi -KioskApps 'com.vendor.a','com.vendor.b' Requires the Microsoft Graph PowerShell SDK and delegated scopes: DeviceManagementServiceConfig.ReadWrite.All, DeviceManagementConfiguration.ReadWrite.All. EXPECTED RESULTS The dedicated enrollment profile and the kiosk device-restrictions profile exist (created if missing). Assign apps (Required-to-device) and the kiosk profile to the device group per the Implementation Guide. Multi-app also requires the Managed Home Screen app assigned to the group. .NOTES Android category — Android dedicated (COSU)/kiosk. Default -WhatIf. Verify the current schema/portal against Microsoft guidance before -Apply. > [CmdletBinding(SupportsShouldProcess=$true)] param([string]$EnrollmentProfileName = 'Android Dedicated - COSU - Pilot', [int]$TokenValidityDays = 90, [string]$KioskProfileName = 'Android Dedicated - Kiosk - Pilot', [ValidateSet('Single','Multi')] [string]$KioskMode = 'Single', [string[]]$KioskApps = @(), [switch]$Apply)

COSU-Set-ScopeTargeting.ps1

Purpose: Creates the Entra DYNAMIC DEVICE group this solution targets policy and apps at — dedicated (COSU) devices have no user, so everything is assigned to a device group whose membership rule matches the enrollment profile name. Idempotent: skips the group if it already exists. Also reports the group's current member count so you can confirm devices are landing in scope. Default -WhatIf; pass -Apply to make changes. HOW TO RUN Preview (default): powershell -ExecutionPolicy Bypass -File .\COSU-Set-ScopeTargeting.ps1 Apply: ... -Apply Custom names: ... -Apply -DeviceGroup 'Android Dedicated - Devices - Prod' -EnrollmentProfileName 'Android Dedicated - COSU - Prod' Requires the Microsoft Graph PowerShell SDK and delegated scope: Group.ReadWrite.All. EXPECTED RESULTS The

dynamic device group exists with the membership rule (device.enrollmentProfileName -eq "<profile>") and processing enabled; its member count is reported. Assign the kiosk profile and managed Google Play apps (Required) to this group next. .NOTES Android category — Android dedicated (COSU)/kiosk. Default -WhatIf. Dynamic device group needs Entra ID P1+. > [CmdletBinding(SupportsShouldProcess=$true)] param([string]$DeviceGroup = 'Android Dedicated - Devices - Pilot', [string]$EnrollmentProfileName = 'Android Dedicated - COSU - Pilot', [switch]$Apply)

How to run: Preview (default): powershell -ExecutionPolicy Bypass -File .\COSU-Set-ScopeTargeting.ps1 Apply: ... -Apply Custom names: ... -Apply -DeviceGroup 'Android Dedicated - Devices - Prod' -EnrollmentProfileName 'Android Dedicated - COSU - Prod' Requires the Microsoft Graph PowerShell SDK and delegated scope: Group.ReadWrite.All. EXPECTED RESULTS The dynamic device group exists with the membership rule (device.enrollmentProfileName -eq "<profile>") and processing enabled; its member count is reported. Assign the kiosk profile and managed Google Play apps (Required) to this group next. .NOTES Android category — Android dedicated (COSU)/kiosk. Default -WhatIf. Dynamic device group needs Entra ID P1+. > [CmdletBinding(SupportsShouldProcess=$true)] param([string]$DeviceGroup = 'Android Dedicated - Devices - Pilot', [string]$EnrollmentProfileName = 'Android Dedicated - COSU - Pilot', [switch]$Apply)

COSU-Validation.ps1

Purpose: Validates that the kiosk is coherent: it reads the kiosk device-restrictions profile's allow-list (kioskModeApps) and the managed Google Play apps that are assigned (Required) to the dedicated DEVICE group, then reports mismatches — apps named in the kiosk but NOT assigned (they will not appear on the device) and apps assigned but NOT in the kiosk (exposed/uncurated). For multi-app, checks that the Managed Home Screen launcher is assigned. Read-only. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\COSU-Validation.ps1 ... -KioskProfileName 'Android Dedicated - Kiosk - Pilot' -DeviceGroup 'Android Dedicated - Devices - Pilot' Requires the Microsoft Graph PowerShell SDK and delegated scopes: DeviceManagementConfiguration.Read.All, DeviceManagementApps.Read.All, Group.Read.All. EXPECTED RESULTS The kiosk allow-list, the apps assigned Required to the device group, and a PASS/FAIL on whether they match (only allowed apps). Any mismatch is listed with the fix. Read-only. .NOTES Android category — Android dedicated (COSU)/kiosk. Read-only. Kiosk schema evolves — verify property names against current Microsoft guidance. > [CmdletBinding()] param([string]$KioskProfileName = 'Android Dedicated - Kiosk - Pilot', [string]$DeviceGroup = 'Android Dedicated - Devices - Pilot', [string]$ManagedHomeScreenPackage = 'com.microsoft.launcher.enterprise')

How to run: powershell -ExecutionPolicy Bypass -File .\COSU-Validation.ps1 ... -KioskProfileName 'Android Dedicated - Kiosk - Pilot' -DeviceGroup 'Android Dedicated - Devices - Pilot' Requires the Microsoft Graph PowerShell SDK and delegated scopes: DeviceManagementConfiguration.Read.All, DeviceManagementApps.Read.All, Group.Read.All. EXPECTED RESULTS The kiosk allow-list, the apps assigned Required to the device group, and a PASS/FAIL on whether they match (only allowed apps). Any mismatch is listed with the fix. Read-only. .NOTES Android category — Android dedicated (COSU)/kiosk. Read-only. Kiosk schema evolves — verify property names against current Microsoft guidance. > [CmdletBinding()] param([string]$KioskProfileName = 'Android Dedicated - Kiosk - Pilot', [string]$DeviceGroup = 'Android Dedicated - Devices - Pilot', [string]$ManagedHomeScreenPackage = 'com.microsoft.launcher.enterprise')